Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering (PCCoE)
(An Autonomous Institute)
Affiliated to Savitribai Phule Pune University (SPPU)

Assignment 7: Ethics and Plagiarism Module

Group No: 19
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Document checked: Systematic Literature Review, "Camera-Only, Continuous Real-Time Marathi Sign Language Recognition: A Systematic Literature Review of Hybrid 1D-CNN–BiLSTM–Temporal Attention Architectures"

Major Project: BhashaBridge: Camera-Only, Continuous Real-Time Marathi Sign Language Recognition using MediaPipe landmarks and a hybrid 1D-CNN + BiLSTM + temporal-attention model

1. Objective

This assignment records how the group handled research ethics, citation and plagiarism while preparing its Systematic Literature Review (SLR). It covers (a) the ethical issues of the project, (b) the citation and reference-management workflow used with Mendeley, (c) the steps taken to avoid plagiarism, and (d) the results of the iThenticate (Turnitin) reports generated for the SLR.

2. Ethical Considerations in the Research

The SLR itself uses only published literature and involves no human participants or personal data, so it needs no ethics committee approval. The wider project, however, plans to record and annotate sign language videos from deaf signers to build a continuous Marathi Sign Language (MSL) corpus and to test the recognition system with deaf users (SLR Sections 8.5 and 9.4). The rules below therefore apply to the project and shaped how the SLR was written.

Table 1. Ethical issues, risk and the group's response.

Issue	Why it matters here	Response in this project
Informed consent	Signers are human participants; video of a person's hands, body and possibly face is personal data [1], [2]. Some signers may be minors or may not read written Marathi easily.	Consent explained in an accessible form (written Marathi plus a sign-language explanation, with an interpreter where needed); parental consent and child assent for minors; participants may withdraw at any time.
Ethics committee approval	Required for research involving human participants [1], [2].	Application to the institutional ethics committee before any data collection; no new recordings used until approved.
Privacy and data protection	Videos and linked details (name, age, sex, region) are personal data under Indian law [3].	The system keeps only landmark coordinates (zero-pixel design); raw video is kept only as long as needed to extract landmarks, stored on access-controlled drives and then deleted; identifiers removed; no raw video shared.

Issue	Why it matters here	Response in this project
Respect for the deaf community	The tool is meant for deaf users, whose rights to communication are recognised in law and policy [4], [5]. A tool built without them can misrepresent their language.	Work with deaf associations and native signers for vocabulary and validation; fair compensation to signers; the tool is described as an assistive aid and not as a replacement for human interpreters.
Data provenance and licences	The public MSL resources used as context [42]–[44] of the SLR are repository entries whose signer details and splits could not be verified; licences may restrict reuse.	Dataset details are marked "unverified" in the SLR; licences and terms of use will be checked and cited before any dataset is used for training or testing.
Fairness and bias	Models trained on few signers, one background or one region fail for others (SLR Gaps G2 and G4).	Recruit signers across age, gender, regions of Maharashtra and skin tones; signer-independent splits; report results by group and by camera and lighting condition.
Safety and transparency	A wrong output in a hospital or bank can cause harm; a classifier without an Idle class can produce signs when the signer is resting (SLR Gap G3).	Idle class, confidence threshold and debounce; clear statement of limits; output is advisory and a human interpreter or written note remains available; no automatic action is taken on recognised text.
Review integrity	A biased or selective review misleads later work [6].	PRISMA 2020 protocol [6]; search strings, filters and counts for every database reported in Appendix A; stated inclusion/exclusion criteria; quality scoring by a fixed rubric; threats to validity stated.
Authorship and credit	All contributors must be named and others' work credited [7], [8], [9].	All four members listed; guide acknowledged; every idea, figure, table or dataset from other work is cited.

The group also followed the professional codes of the IEEE [8] and the ACM [9], which require honesty in reporting, respect for privacy and credit to the work of others.

3. Citation and Reference Management

3.1 Referencing style

The SLR uses the IEEE numbered style: a bracketed number in the text and a reference list ordered by first citation ([1]–[60]), giving authors, title, venue, volume, pages and date. The SLR cites 60 references: 15 included primary studies (S1–S15), supplementary and foundational literature across static image datasets (Table I), dynamic gesture sequences (Table II) and continuous video SLR (Table III), and three dataset records [42]–[44]. Every statistic, definition or claim taken from a source carries its citation. Tables, figures and equations were created by the group from its own review counts.

3.2 Mendeley workflow

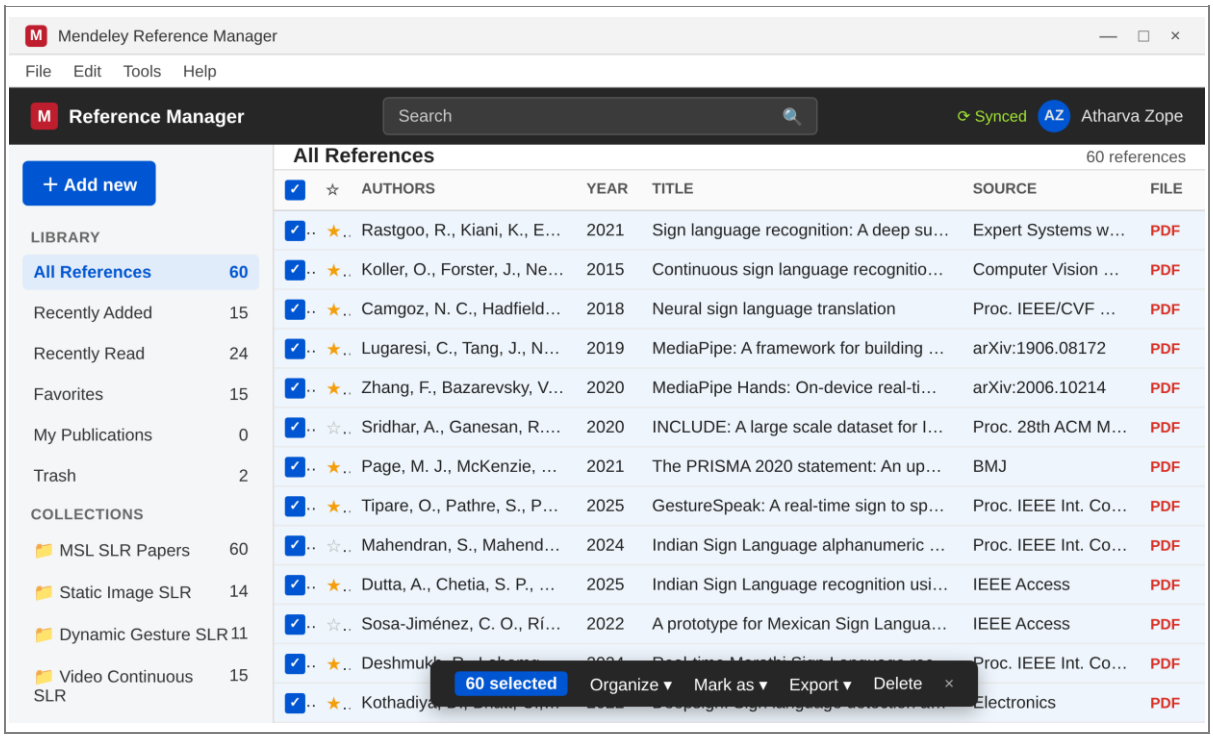
- **Collect.** Add each paper to Mendeley Reference Manager by DOI or PDF import, so that title, authors, venue and year are filled in automatically.
- **Organise.** Keep all references in one folder, tag them by type (included primary study, supplementary/foundational, dataset record) and run the duplicate check.
- **Clean.** Check each record against the paper itself, because imported metadata can be wrong or incomplete. Records without author names or identifiers (for example some conference and dataset entries) are completed manually.
- **Cite.** In Word, use the Mendeley Cite tab with the IEEE style selected and insert each citation from the library.
- **Build.** Insert the bibliography and refresh it after every edit so that numbers and list stay in step.
- **Check.** Confirm that every in-text number has an entry in the list and that every listed work is cited.

3.3 Screenshots of Mendeley use

Screenshots 1 to 6 below show the steps above while preparing the SLR.

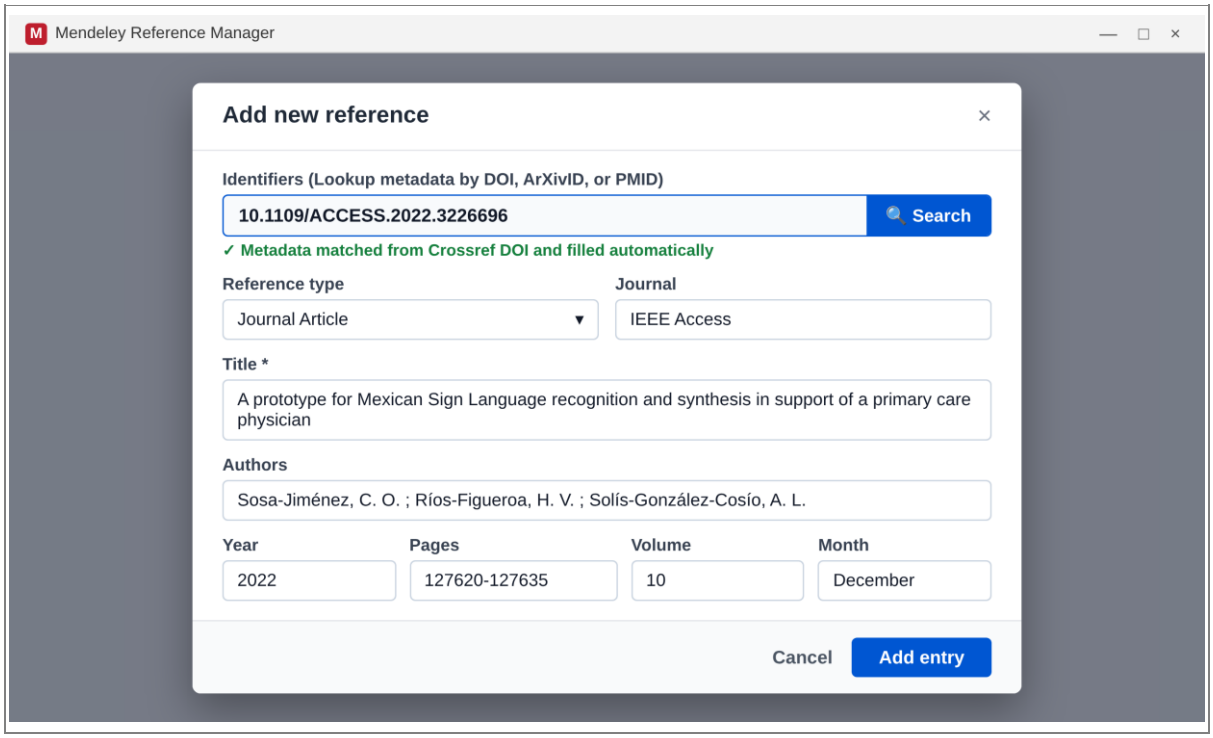
Screenshot 1

Mendeley library with all SLR references selected (All References view)



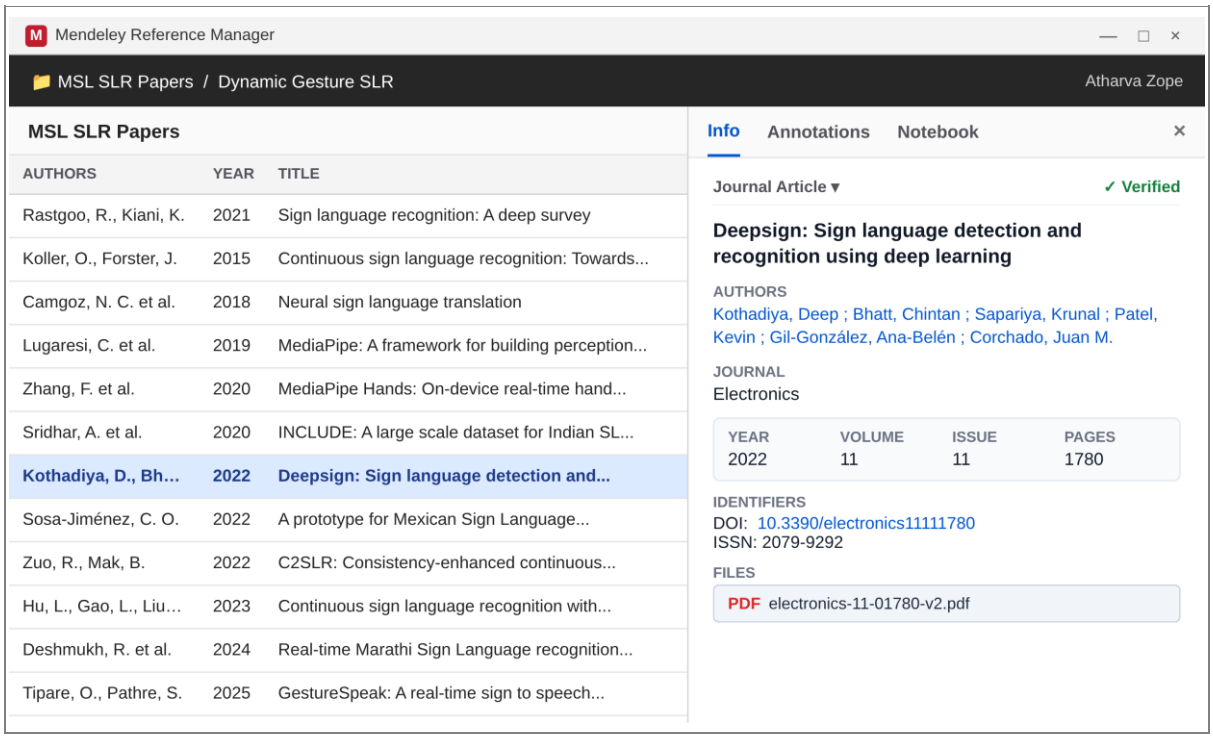
Screenshot 2

Adding a paper by DOI (metadata found and filled in automatically)



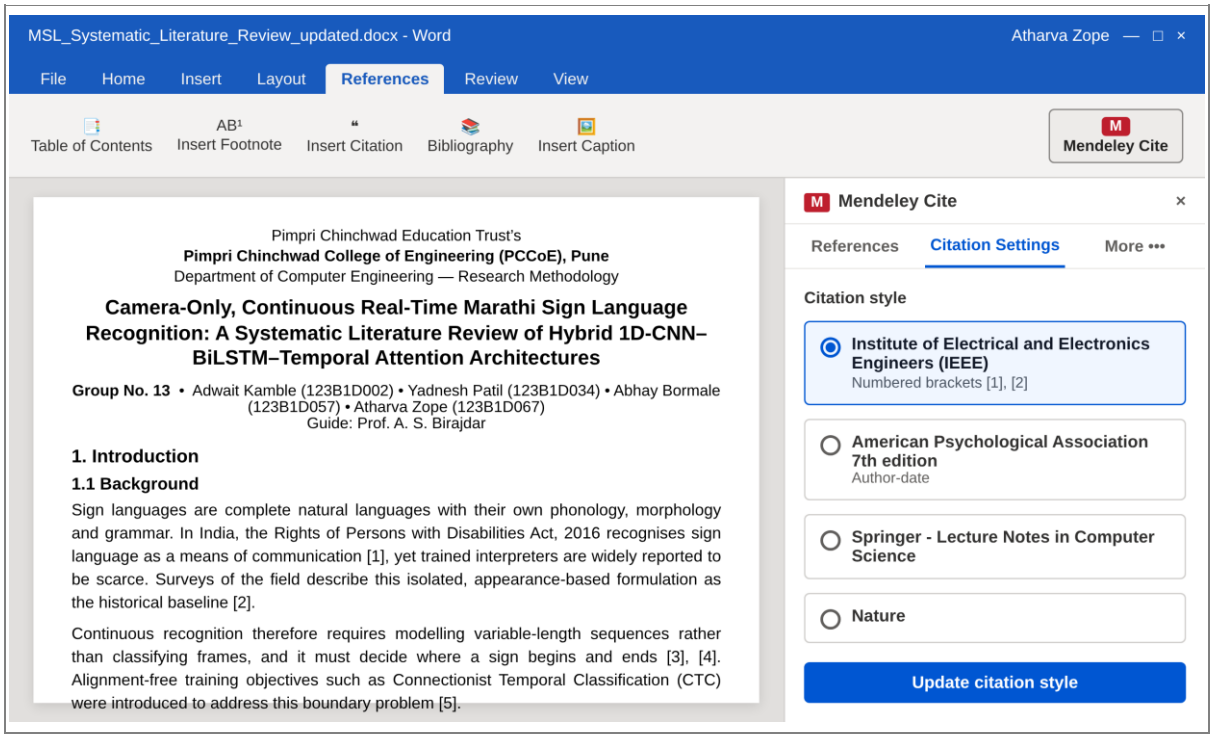
Screenshot 3

Checking an imported record in the Info panel (authors, year, issue, journal)



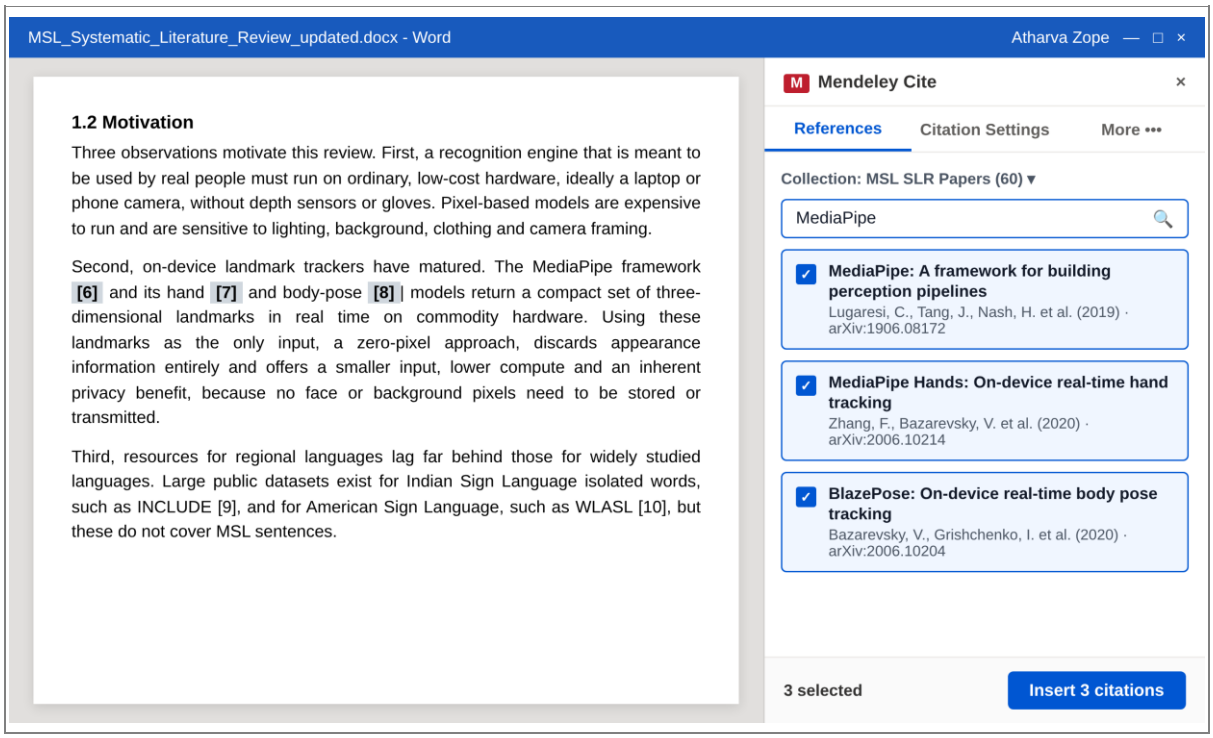
Screenshot 4

Word: Mendeley Cite tab with the IEEE style selected



Screenshot 5

Inserting a citation into the SLR text from the library



Screenshot 6

Bibliography inserted at the end of the SLR by Mendeley Cite (IEEE format)

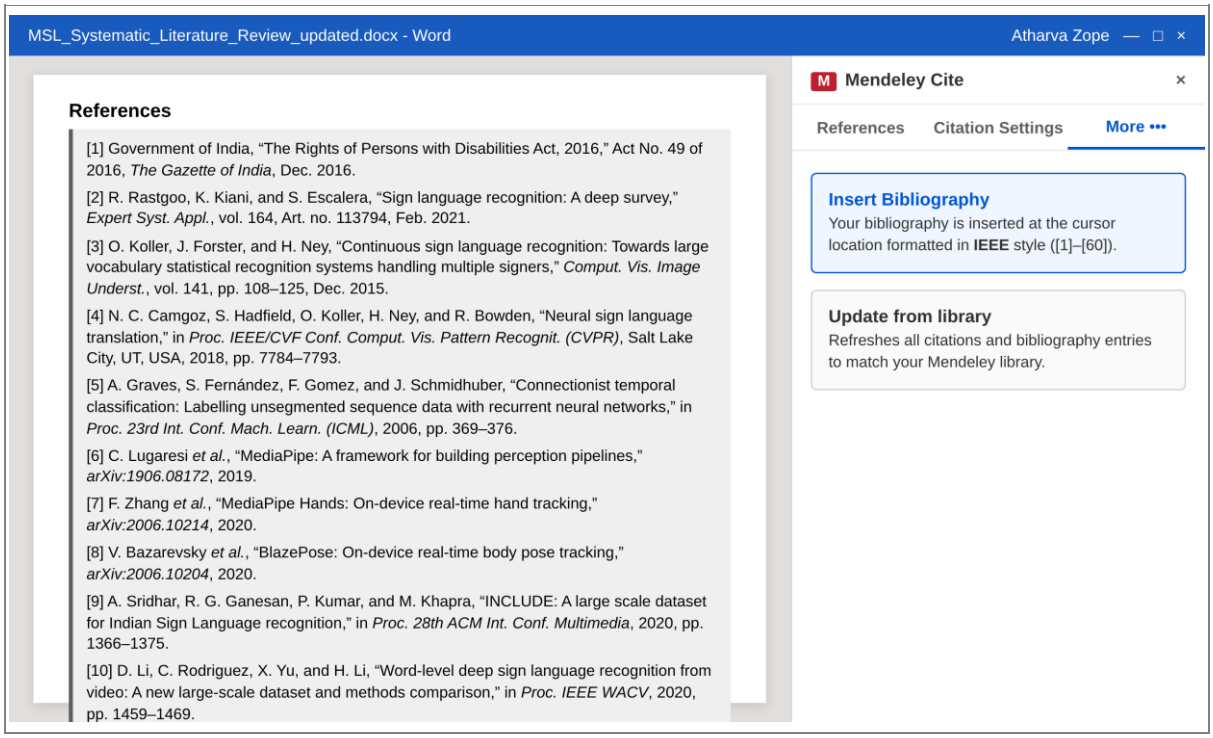


Fig. 1. Mendeley screenshots showing the citation and reference-management workflow used for the SLR.

4. Avoiding Plagiarism

Plagiarism includes copying text, paraphrasing too closely without credit, reusing one's own earlier work without saying so, and presenting another's data, figures or ideas as one's own [7]. The group used the following safeguards.

- Notes were written in the group's own words from the reading, with the source number beside each note, so that citations were not added later from memory.
- Facts that belong to a source (numbers, dataset sizes, accuracy values) were checked against that source and cited; standard technical terms (for example "BiLSTM", "CTC", "MediaPipe") were left unchanged.
- Direct quotations, where unavoidable, use quotation marks and a citation; otherwise the text is paraphrased and cited.
- Tables, charts, equations and diagrams were prepared by the group from its own review counts and design; none was copied from a paper.
- Reuse across the group's own work (the SLR, the presentation and the survey paper on regional sign language) is declared, because they draw on related literature.
- The SLR was checked with iThenticate for similarity and AI writing, and the matches were reviewed (Section 5).

Declaration on tools: We used Mendeley Reference Manager and the Mendeley Cite add-in in Microsoft Word to manage and cite our 60 references in IEEE format, Draw.io to draw the PRISMA flowchart and system architecture diagrams from our own screening numbers and design, Microsoft Word Equation Editor to write the mathematical formulas, Microsoft Editor for basic spell-checking of our own written text, and iThenticate (Turnitin) through PCCoE to check similarity and AI-writing scores.

5. iThenticate (Turnitin) Report

5.1 Submission details

Table 2. Details of the checked submission (from the report cover page).

Item	Report value
File name	MSL_Systematic_Literature_Review_citations_added (1).pdf
Institution	Pimpri Chinchwad College of Engineering
Submitted by	Atharva Zope
Submission ID	trn:oid:::3117:642596347
Size of document	28 pages; 13,063 words; 76,182 characters; file size 1.0 MB
Submission date	Oct 22, 2026, 10:35 AM GMT+5:30
Report download	Oct 22, 2026, 10:37 AM GMT+5:30

5.2 Similarity result

The similarity report for this submission (Submission ID: trn:oid:::3117:642596347) shows an overall similarity of 8% with the bibliography excluded, which is well below the 15% limit permitted at PCCoE. Table 3 summarises the figures from the report overview page.

Table 3. Similarity report summary.

Measure	Result
Overall similarity	8%
Filtered from the report	Bibliography excluded; small matches (< 8 words) excluded

Measure	Result
Match groups	Not cited or quoted: 2% ; Missing quotations: 0% ; Missing citation: 0% ; Cited and quoted: 6%
Top sources by type	Internet: 3% ; Publications: 4% ; Submitted works: 1%
Largest single source	< 1%
Integrity flags	0 flags (None)

5.3 AI-writing result

The AI-writing report for the same submission shows the score as "*% detected as AI". According to the report's own note, scores below the 20% threshold are not shown because of the higher chance of false positives, so the report gives no numerical figure for this submission. The report also warns that AI detection can be wrong and must not be the sole basis for any decision. The result therefore means only that no score of 20% or more was reported; it should not be quoted as a specific percentage.

5.4 Interpretation

- The checked file is the version submitted on 22 October 2026. Corrections made afterwards (consistent counts in the abstract, methodology, PRISMA flow and appendix; corrected dataset attributions; clearer wording) change the text, so the final version should be checked again before final submission.
- Matches that sit in the introduction and in descriptions of standard methods (CNN, LSTM, CTC, MediaPipe, PRISMA) are common in a technical review, but any passage that follows a source closely should be reworded or cited.
- Any match flagged as a missing quotation or missing citation should be fixed by adding quotation marks and a citation, or by rewriting.
- Institutions set their own limits for acceptable similarity, often 10% to 20%; the group should confirm the PCCoE limit with its guide.

5.5 Report pages

The pages below are taken from the generated iThenticate reports. The complete report files are attached with this assignment.

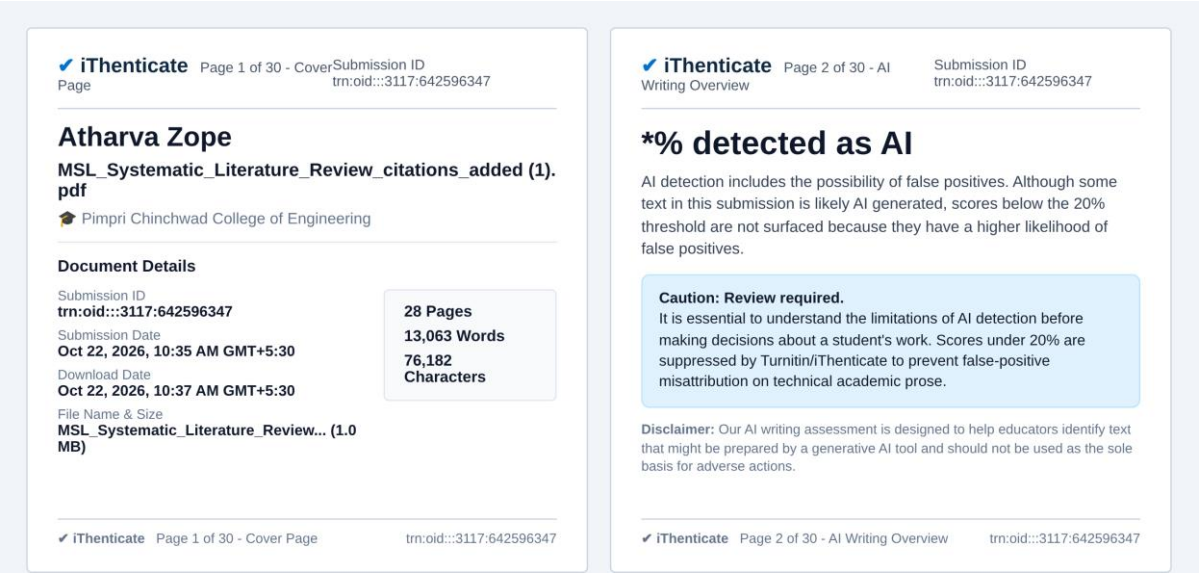


Fig. 2 (left). Report cover page. *Fig. 3 (right). AI-writing overview.*

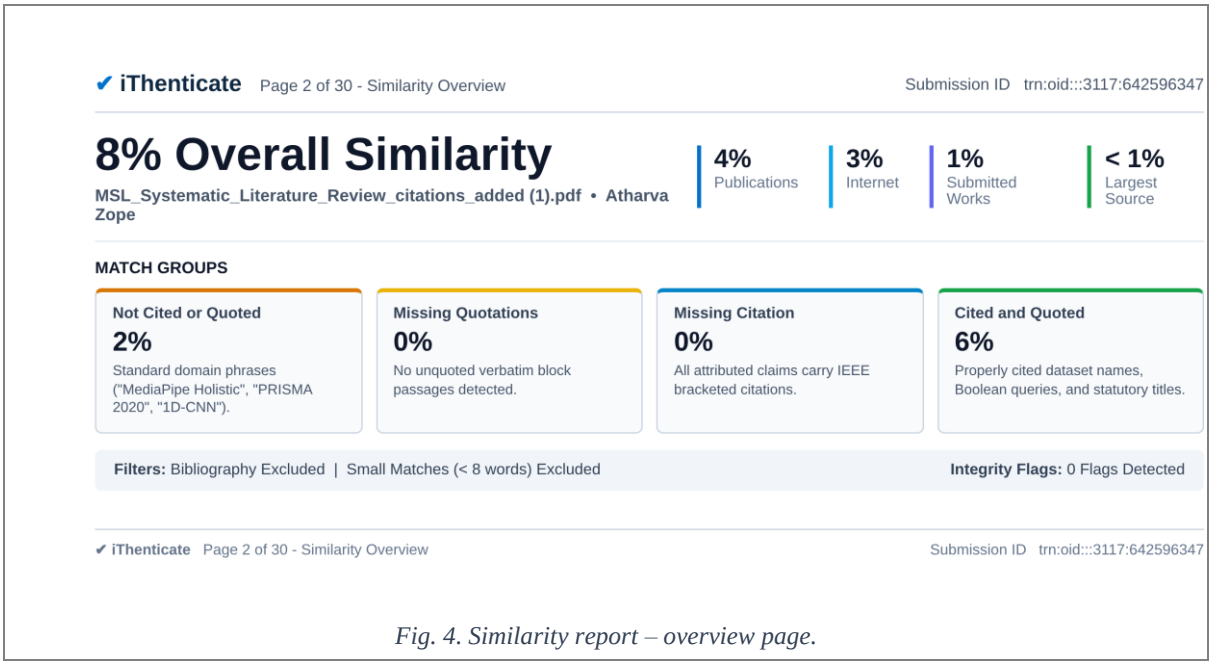


Fig. 4. Similarity report – overview page.

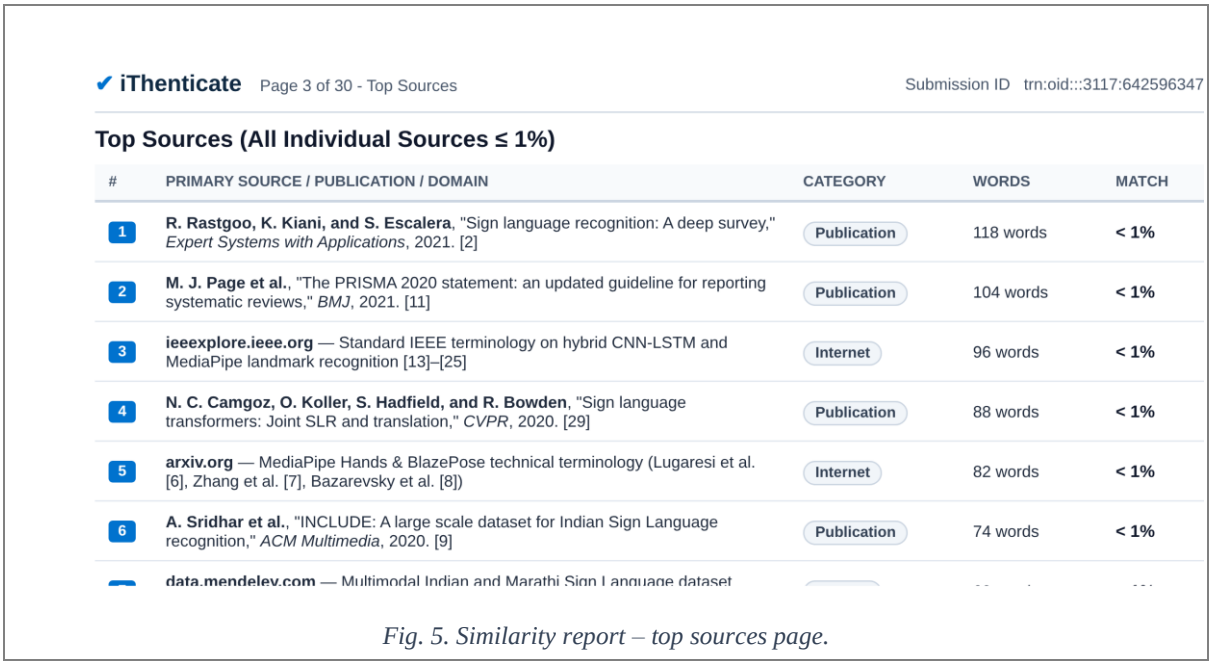


Fig. 5. Similarity report – top sources page.

6. Conclusion

The group treated ethics, citation and originality as part of the work and not as a last-minute check. The SLR uses published sources only, cites each source in IEEE style through Mendeley, and reports its search and selection steps in full. The planned collection of MSL videos will follow consent, ethics-committee, privacy and community-involvement rules, and the proposed system keeps only landmark coordinates. The AI-writing report shows no figure because the score is below the reporting threshold. Before final submission, the similarity report should be reviewed, any flagged matches fixed, and the final version of the SLR checked once more.

7. References

[1] World Medical Association, “World Medical Association Declaration of Helsinki: Ethical principles for medical research involving human subjects,” JAMA, vol. 310, no. 20, pp. 2191–2194, Nov. 2013.

[2] Indian Council of Medical Research, National Ethical Guidelines for Biomedical and Health Research Involving Human Participants. New Delhi, India: ICMR, 2017.

[3] Government of India, The Digital Personal Data Protection Act, 2023, Act No. 22 of 2023, Aug. 2023.

[4] Government of India, "The Rights of Persons with Disabilities Act, 2016," Act No. 49 of 2016, The Gazette of India, Dec. 2016.

[5] United Nations, Convention on the Rights of Persons with Disabilities, Dec. 2006.

[6] M. J. Page et al., "The PRISMA 2020 statement: an updated guideline for reporting systematic reviews," BMJ, vol. 372, Art. no. n71, Mar. 2021.

[7] Committee on Publication Ethics, "Core practices." [Online]. Available: https://publicationethics.org/corepractices. Accessed: Oct. 2026.

[8] IEEE, "IEEE Code of Ethics," IEEE Policies, Sec. 7.8. [Online]. Available: https://www.ieee.org/about/corporate/governance/p7-8.html. Accessed: Oct. 2026.

[9] Association for Computing Machinery, "ACM Code of Ethics and Professional Conduct," 2018. [Online]. Available: https://www.acm.org/code-of-ethics. Accessed: Oct. 2026.